INSTRUCTION MANUAL

MAIZE DEGERMER

MLT- 21 / 26

WARNING

- Do not carry out any operation, maintenance or inspection of the machine until you have read and well understood the contents of this instruction manual.
- Keep this manual at hand whenever any operation, maintenance or inspection of the machine is being carried out.

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1. LIMITED WARRANTY

The machine was produced under quality control and inspection. But, if there are any problems under normal use the limited warranty for the machine extends for the first year beginning on the date of purchase of the machine.

1. Please contact our agency or sales office to repair. If there is a problem within limited warranty period, We will repair for free (only under normal use condition). After the warranty period, repair fee would be billed.
2. It will be charged during limited warranty period following conditions.
   (1) Damages & losses misuse of improper or operated installation with no care.
   (2) Damages & losses unauthorized modification or unauthorized repair.
   (3) Damages & losses because of disasters (fire, flood, storm and earthquake etc.), pollution, unusual voltage & damaged by sea air(rust).
   (4) Damages & losses which you moved or dropped the machine after setting.
   (5) Damages & losses which use of unauthorized spare parts or repair at unauthorized agency or factory.
3. We will supply parts of the machine for 7 years after stop of production. But within this period some special parts are exceptional.
4. Basically, We will stop to supply parts in 7 years after discontinued to produce. But if you have a request to order the parts, We will discuss about it. All warranty informations, machine features and specifications are subject to change without notice.

2. SAFETY

WARNING

Do not carry out any operation, maintenance or inspection until you have read and well understood the contents of this instruction manual.

- Keep this manual at hand for reference whenever any operation, maintenance or inspection of the machine is being carried out.
- When you operate, maintain or inspect the machine, you are requested to follow instructions and warnings as stipulated in this manual. If you have any questions or indistinct points, you should suspend operation of the machine or your work until you can clarify it with the Satake office listed at the end of this manual for inquiries about the machine.
- Degpa SA shall not be responsible for any injury, loss or damage caused by failure to observe instructions stipulated in this manual, by misuse or modification without our permission.

This chapter describes the definition of hazard signal words as well as safety precautions to be observed at the time of operation, maintenance or inspection as detailed and classified in the “General Inspections” and “ Special Precautions.”

2.1 Warning / Caution Types and Meanings

Hazard warning in this instruction manual and warning stickers attached on the machine main body are classified into the following three kinds, depending on the degree of risk of accident.

Operator must recognize the importance of the warnings, and carefully follow the instructions given in this manual.

 WARNING	This shows potential hazard. If you do not follow this, there is some possibility of fatal or serious injury.
 CAUTION	This shows potential hazard. If you do not follow this, there is some possibility of middle or minor class injury, damages of the equipment / machinery or misconduct in material or product.
 NOTE	This is used for calling special attention to or adding emphasis to a certain information required to be noticed.

2.2 General Instructions

WARNING

- The operator is supposed to pay attention to the following points to ensure safety in operating, inspecting and serving the machine.

(1) Operator must contain long hair in a hair cover and neatly wear clothes and shoes that are suitable for work. When carrying out any inspection or maintenance, wear appropriate protective gear such as helmet and safety shoes.

(2) Keep clean the surroundings and passages around the machine.

(3) Do not pour water over the machine. Water in the machine shorts the electric circuits, consequently causing the machine to break down.

(4) Do not fail to ground the machine in order to protect a person from being involved in electric accidents and to prevent fires from occurring due to current leaks.

(5) Never touch the live parts inside the machine after turning on power.

(6) Do not permit any person who has not been properly trained to operate or service the machine.

(7) Keep out any person other than those permitted by the operator. Never let children come close to the machine.

(8) Do not fail to inspect the machine before starting operation. When carrying out any inspection work, always power off and clearly indicate that the machine is under inspection or maintenance both in the control room and control panel at the work shop.

(9) Always power off before inspection and maintenance and follow the appropriate lock out procedure. After the inspection, make sure that there are no tools left.

(10) Check for any looseness or damage on bolts, nuts and belts. Make sure to restore the cover after the checking.

(11) Check for any damages on power cords and wire cables. Also, check for any looseness or disconnection of connectors and plugs.

(12) Operate the machine with all covers attached. Do not attempt to remove any cover during the machine operation.

(13) Always stop the machine before greasing the driving part of the machine.

(14) Operator should well understand the power off procedure for a case of emergency.

(15) When working with two or more persons, each person must confirm communication with others before starting work.

2.3 Special Precautions

(1) Install the machine in a level and stable place. Unleveled and unstable installation will cause vibration and poor performance of the machine.

(2) Do not operate, inspect or maintain the machine with any part that is not specified in this instruction manual.

(3) Keep sufficient working space around the machine.

(4) Before turning the power on, confirm the safety of the surroundings and passages around the machine.

(5) Handle the machine with care. Careless handling of the machine may lead to malfunction and failure of the machine.

(6) Keep alert to the machine, If any abnormality is found, stop the operation and check the machine to ensure safe operation.

(7) When using a ladder, it must be set in stable location.

(8) In case the machine is used in a dusty place, wear a dust-proof mask and goggles.

(9) If employees of a plant are unable to read or write, the owner of the plant must explicitly draw their attention to any existing hazards and must give them special instruction.

2.4 Warning Stickers

NOTE

How to handle Warning Stickers

- Check if letters and illustration on the sticker are legible and if not, remove dirt on the sticker or replace it.
- Use cloth, water or cleanser to remove dirt. Do not use organic solvent or gasoline.
- In case the sticker is damaged, lost, or illegible, replace it with a new one.

This machine is provided with warning stickers on certain locations where a special precaution is required. It is important for operators to know the location and contents of each stickers and follow the instruction on the sticker to prevent accidents.

2.5 Location of Warning Stickers

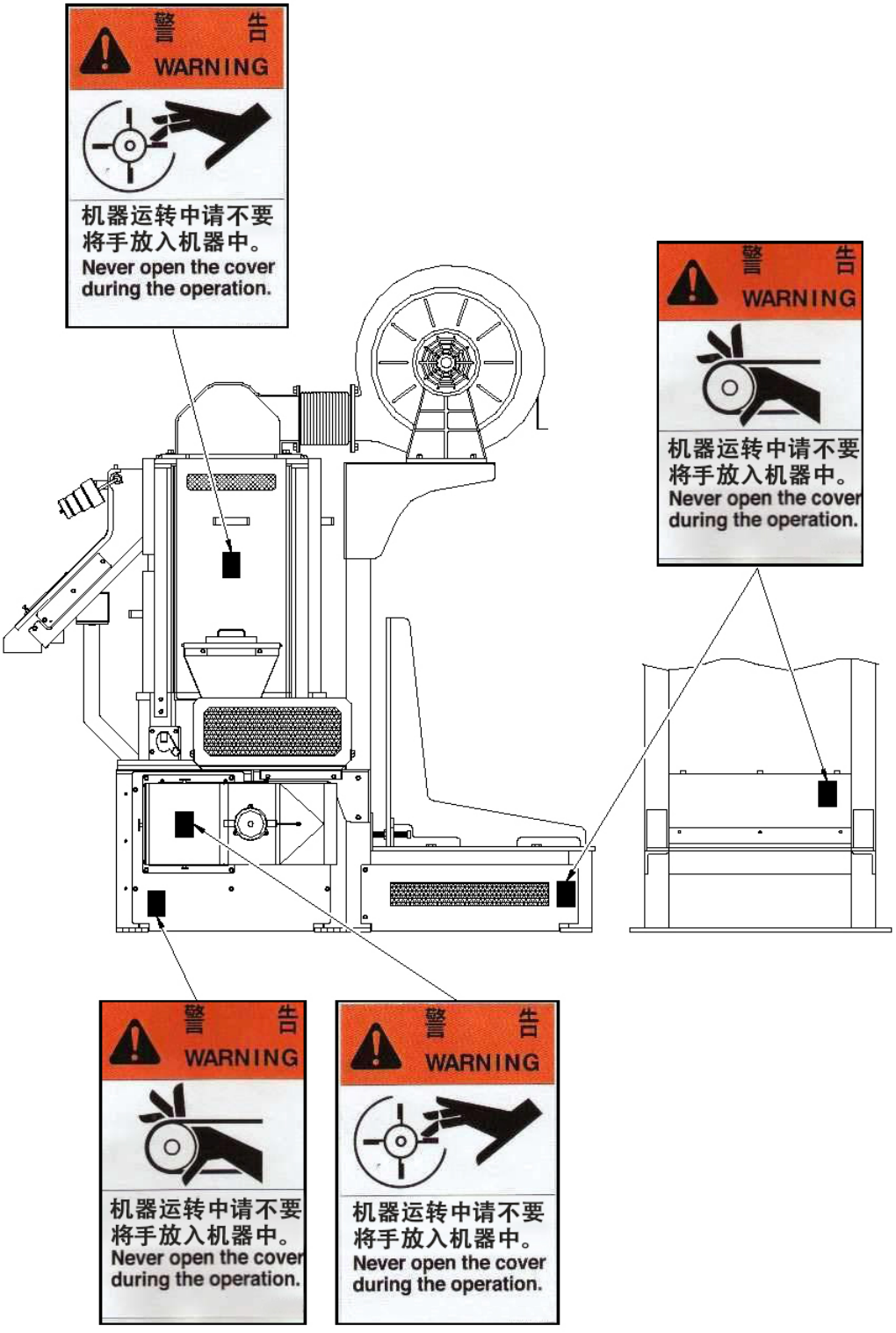


Fig. 2-1 Right Side View

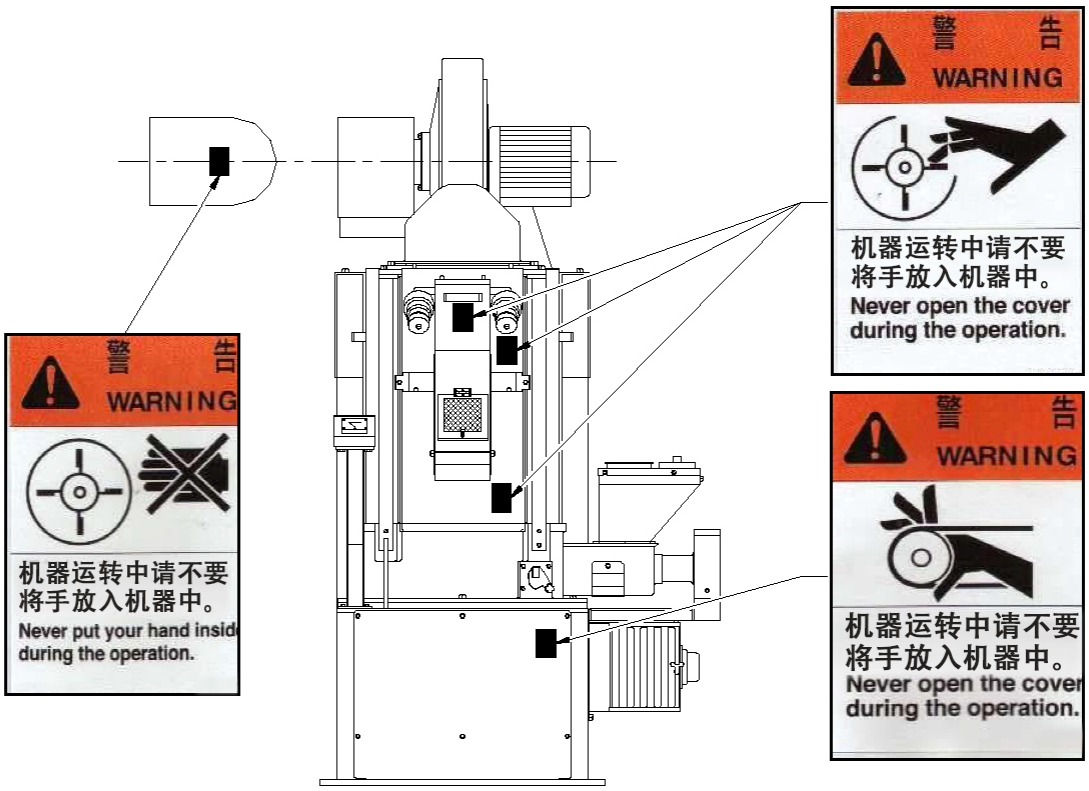


Fig. 2-2 Front View

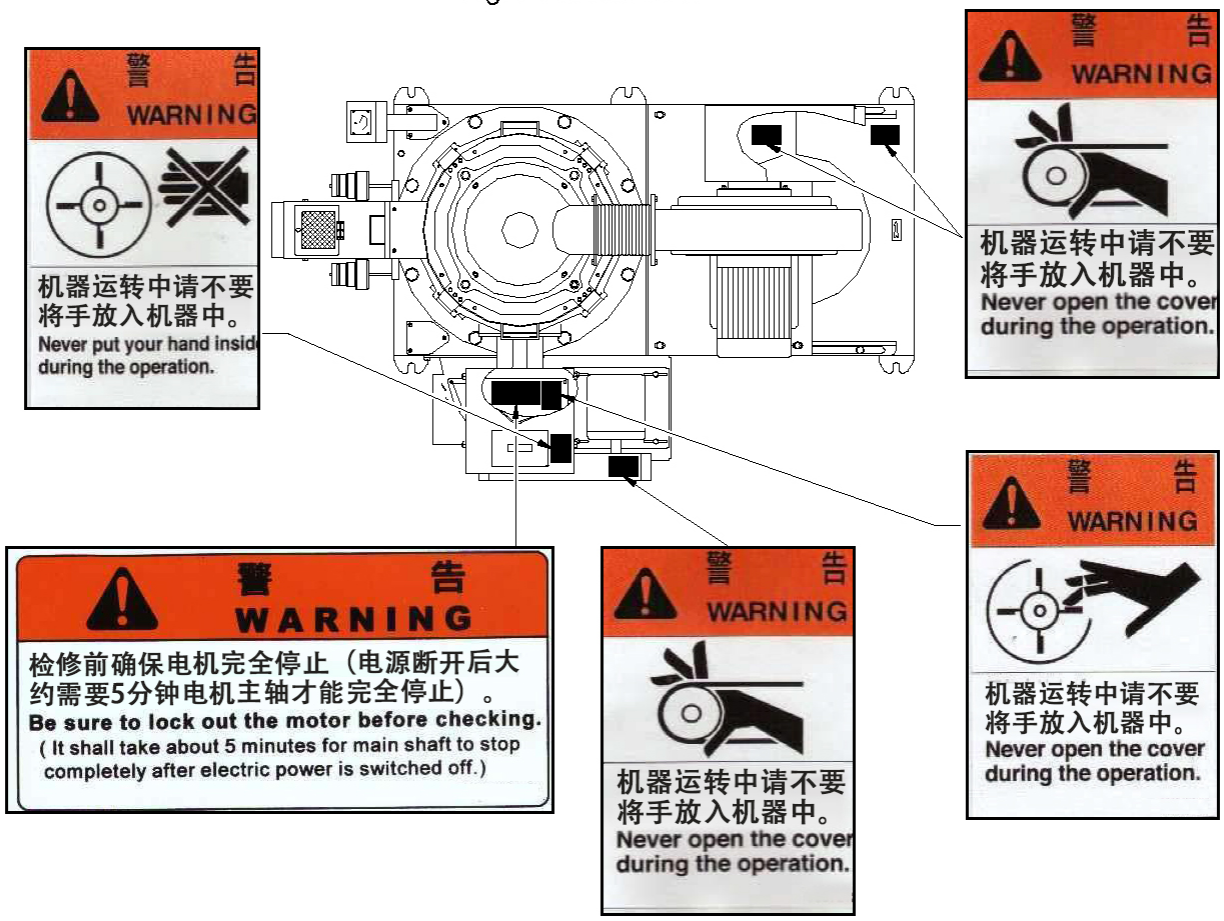


Fig. 2-3 Top View

3. SPECIFICATIONS

Model			MLT
Input Capacity (T/H)			4 ~ 6 tons/hr.
Required Power			45 ~ 55kW, 6P
Suction Air Volume (m^3/min)			45
Static Pressure (kPa)			-1 ~ -1.5
Size of Bran Suction Duct			8 inch Dia.
Revolution of Main shaft	Without Mist	50 Hz	465 rpm
		60 Hz	485 rpm

※ Input capacity is varied. Depends on material and configuration of whitener(s).

4. DIMENSIONS

Unit:mm

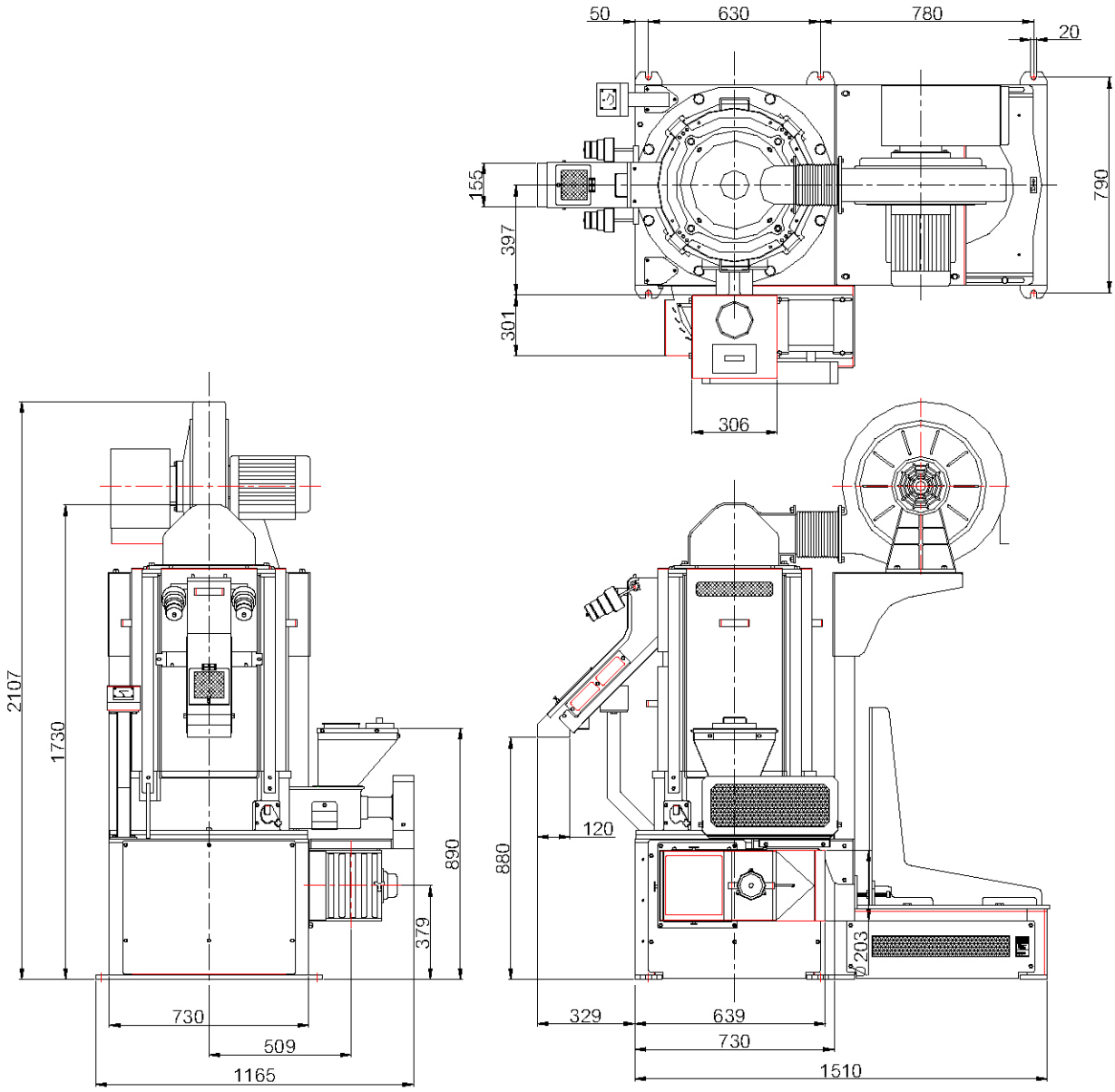


Fig. 4-1

5. CONSTRUCTION

5.1 Names of Components

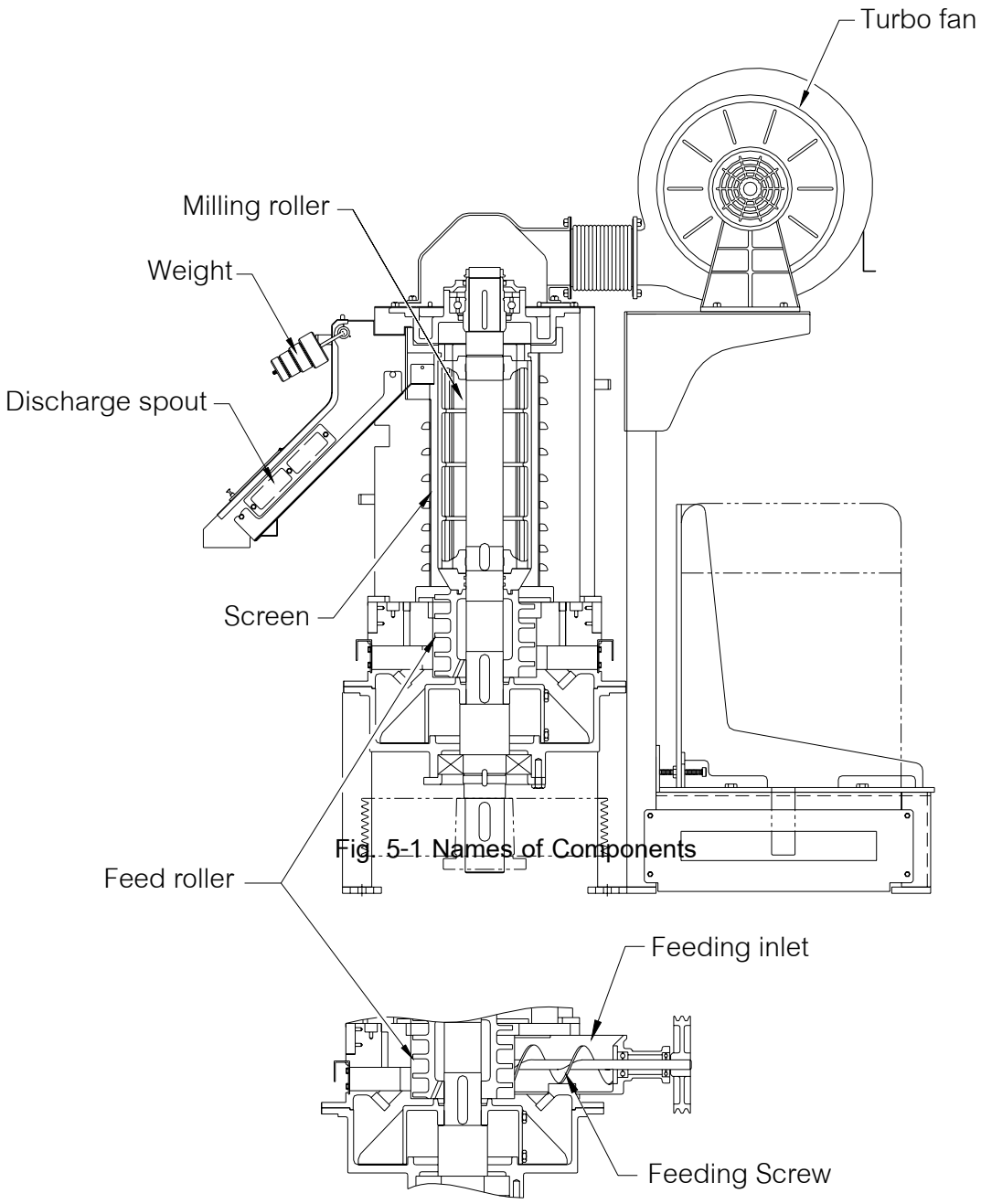


Fig. 5-1 Names of Components

No.	Name	Function
1	Feeding inlet	To feed material.
2	Feeding screw	To transfer material from feeding inlet into machine inside.
3	Feeding roller	To lead material into milling chamber.
4	Milling roller	Equipped with 3 pieces of resistant plates and 3 air injection outlets to mill material.
5	Screen	To separate white rice by passing bran from the slit of screen.
6	Discharge spout	To discharge milled material to the outside of machine.
7	Weight	To adjust the milling degree by changing the position, quantity and weight.
8	Turbo fan	To pass the bran through screen by injecting the air from the injection holes of milling roller into milling chamber.

Table 5-1 Names and Function of Components

6. INSTALLATION

6.1 Rigging Method of the Machine

WARNING

- Use a wire having sufficient strength because the machine weight without motor is about 1200 kgs.

- Use the four eye bolts attached to the machine when transferring or installing the machine.
- Use the great care not to damage the side cover etc. when carrying out rigging work.

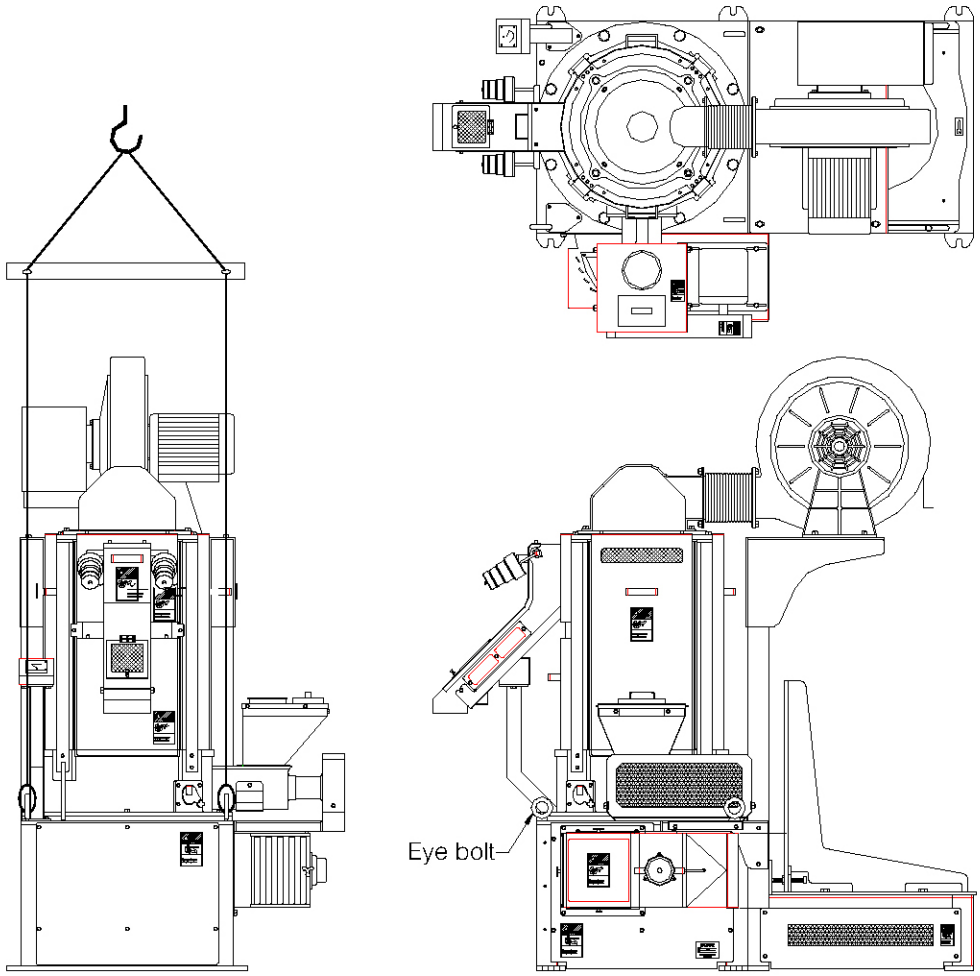


Fig. 6-1

6.2 Confirmation of Levelness

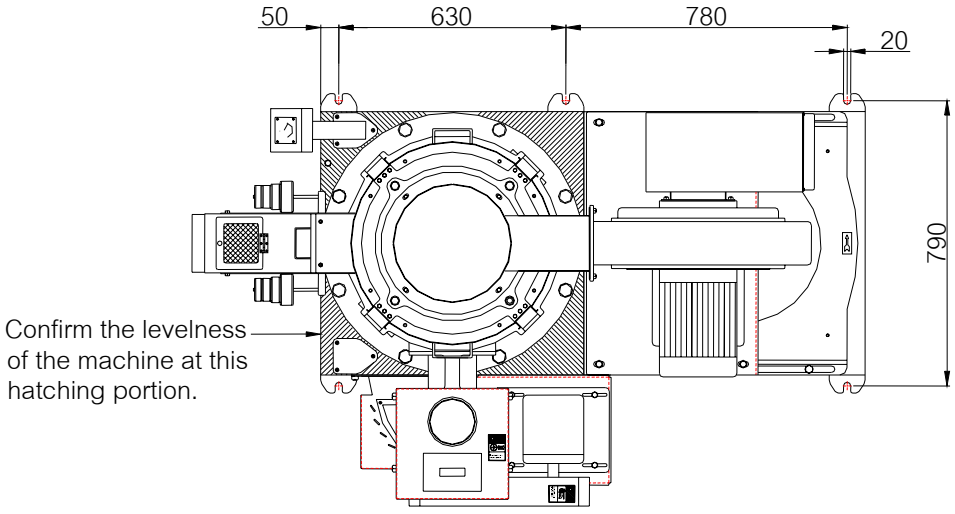
CAUTION

- If the machine is not set horizontally, it may not be able to exhibit its expected capability fully.
- Ensure the levelness of the machine and bolt it securely to the floor with bolts of more than M16.

Set the machine at the specified position and make sure that it is placed horizontally. If not, insert a liner at the position of bolt to adjust the levelness of the machine. Then bolt the machine securely to the floor with bolts.

6.3 Position Setting

Bolt the machine securely to the floor with M16 bolts at 6 pieces of the fixing plates of bolts.



6.4 Selection of Floor

As the machine without motor weighs about 1200 kgs. The floor to be selected for installation must have enough strength.

6.5 Maintenance Space

As installation, provide enough space around the machine for servicing or access as shown in **Fig. 6-3** (The dimensions indicated are minimum.)

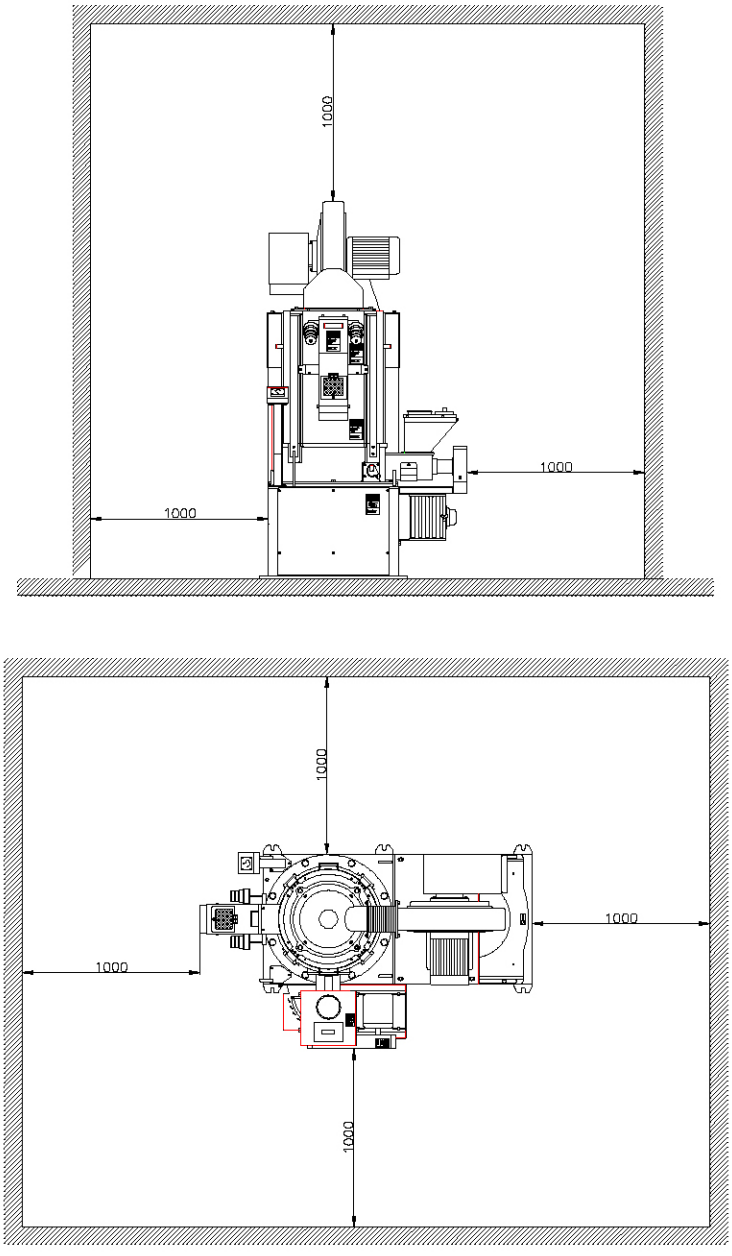


Fig. 6-3

6.6 Component Parts of Weight and Installation

WARNING

- In case that the weight is easy to move on the control level, it may drop from the control level when assembling it into the control level and setting in position, assemble with care.

6.6.1 Component parts or weight

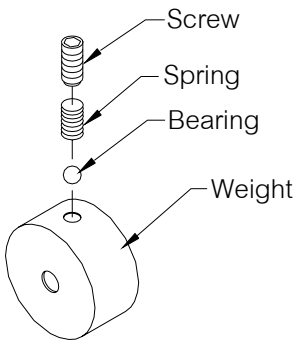


Fig. 6-4

6.6.2 Weight Installation

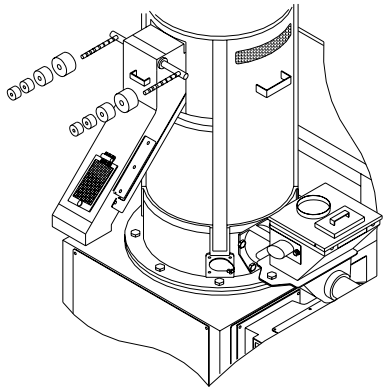


Fig. 6-5

Reassemble weight assembly

- Fixed two levers on shaft as shown in **Fig. 6-5**
- Install weight on lever.

Adjust spring as follow:

1) Turn the screw clockwise to make the spring stronger, using a minus-screw driver.
2) Slide the weight in the control lever the check the strength of spring.
3) Turn the screw clockwise further when it is not strong enough, and counterclockwise when it is too strong.

6.7 Fitting the Motor

Make an opening on the safety cover when you fix the motor at site.

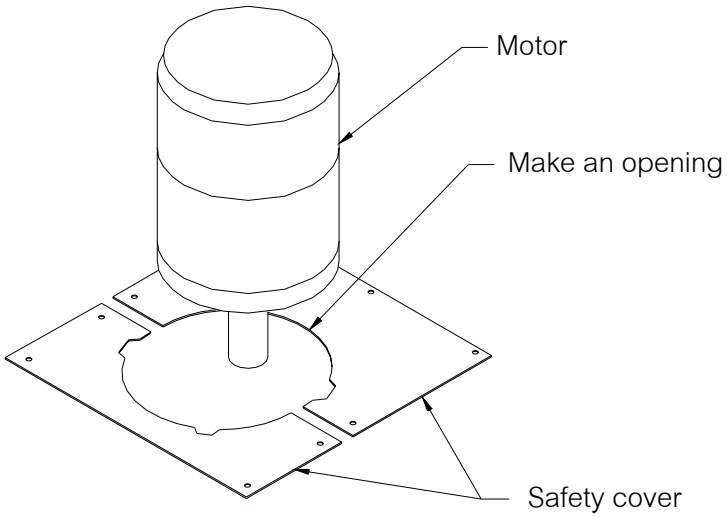


Fig. 6-6

7. PREPARATION FOR OPERATION

WARNING

- Turn off the power switch and the breaker first before starting examination of the rice milling chamber, and display sign board indicating "**UNDER INSPECTION**".
- Turn the pulley with a hand slowly and examine the internal.

7.1 Inspection of Belt Tension and Milling Chamber

Remove the cover on the lower part of the machine, and conduct the following examination.

(1) Maintain proper tension of V-belt.

Weak tension of V-belt will cause slip, and excessive tension may adversely affect the durability of bearings and the V-belt with resultant failures and accidents on the machine.

(2) Inspect the inside of the rice milling chamber for any foreign matters such as bolts and nuts.

7.2 Inspection in No-Load Operation

Operate the motor of the machine free from load, and check for any excessive heat on the bearings and any extraordinary vibration or noise on the machine.

Make sure the rotating direction of the motor (counterclockwise seen from the top).

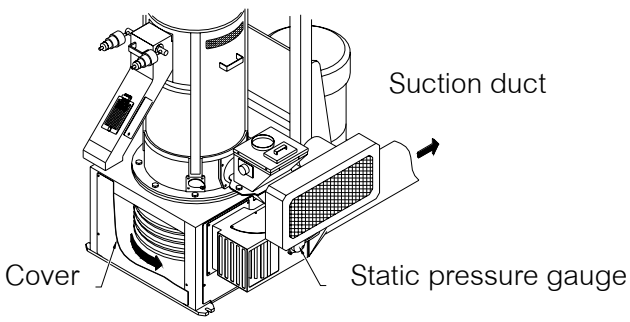


Fig. 7-1 Rotating Direction and Suction Duct

7.3 Belt Tension Adjustment

WARNING

- Turn off the power switch and the breaker at first before adjusting the tension of belts and also display a signboard notifying "UNDER INSPECTION".

(1) Shorten the distance between axes.

Always shorten the distance between axes prior to attaching the belt. Give proper tension to the belt after completion of coupling of the belt.

(2) Maintain proper tension on V-belt.

Belt tension is measured by the amount of load to cause 1.6 mm flexion per 100mm belt span when it is pulled down by a spring scale perpendicularly at the center of the span. Tense the belt so that the flexion load becomes A kg to B kg (C kg in case of renewal of re-tensioning).

The optimum belt tension is such that the belt does not slip against maximum load being given to it and yet the belt is tensioned as loose as possible.
It is important to maintain belt tension between A kg and C kg. Excessive tension exceeding B kg may shorten durability of belts and bearings, while weaker tension than A kg may cause slips.

Table 7-1 Pulley Dia and Load of Flexion

Unit : kg

Dia of motor pulley (mm)	A	B	C
∅180 ~ 230	5.9	8.7	7.6
∅231 ~ 310	7.1	10.6	9.2
∅311 ~ 400	8.4	12.4	10.8

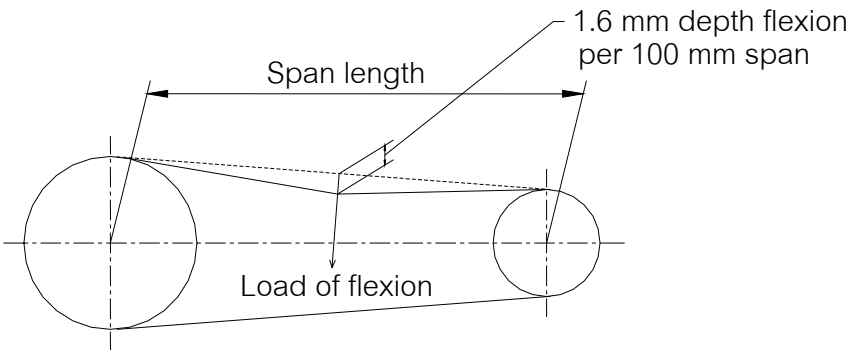


Fig. 7-2 Belt Tension

After a proper tension is given to the belt, run the V-belt for a few minutes and make sure that no excessive looseness of the belt exists.

For a simple test method of belt tension, touch the pulley and the bearing by hand and check their temperatures.

- When the pulley is too hot: The belt tension may be too weak or the belt may be slipping.
- When the bearing part is too hot: The tension of belts may be too tight.

7.4 Air Volume Adjustment

Adjust the air volume so that the static pressure indicated on the static pressure gauge equipped with the machine shall be -0.9 ~ -1.5 kPa (negative pressure).

8. OPERATION

8.1 Operation Procedure

NOTE

- There is a fear of tripping of the machine motor at starting material's flow or at an abrupt adjustment of the weight.
- Ensure to have all the covers restored in position during operation of the machine for securing safety.

(1) Turn on the bran suction fan motor switch.

(2) Turn on the machine motor switch with setting the flow-rate dial to "0".

(3) Operate the machine at no - load and check for unusual vibration or noise.

(4) Press the smallest weight into the control lever, and open the flow rate dial.

(5) The milling chamber becomes full with materials, milled material begins to be discharged, then the flow of electricity stabilizes. After that, adjust the weight while paying attention to the electric current not to exceed the rated value specified.

8.2 Trip of Machine Motor



CAUTION

- Along hour inching or repetitious inching may damage the magnetic contactor of motor. If the motor will not run despite of any inching efforts, power off the motor and dismantle the machine to take out residual materials.
- Ensure to make the inching operation safety by fitting the shutter of residual material discharging outlet.

When the machine motor has tripped and is unable to rotate in spite of attempting re-start up, or when the operator wishes to discharge remaining of materials from the machine, follow the procedure of restoration as described below:

(1) Terminate the machine. (Power off.)

↓

(2) Stop the bran suction fan.

↓

(3) Remove the shutter of the residual material discharging outlet located on the side of the machine and extract residual materials with a vacuum cleaner.

↓

(4) If the materials are blocked in the milling chamber, and it is difficult to take them out, fit the shutter and power on the motor and conduct inching operations to loosen the cluster of materials and vacuum them out.

9. MAINTENANCE AND INSPECTION

WARNING

- Prior to cleaning or inspecting parts, turn off the power switch and the breaker, and display an "**UNDER INSPECTION**" notice.
- Carry out inspection or maintenance work after you confirm a complete stop of the machine. Be careful especially because the main shaft of the machine keeps running for about 5 minutes even after the stop button is depressed.

9.1 Check of Bolts

Periodically check bolts (especially pulley setting bolts) for any loosening.

9.2 Check of Belts

Periodically inspect belt tension.

(If the belt retreats 10mm when lightly pressed with fingers, it is considered normal.)

9.3 Check of Bran Adhering Plate

Periodically clean the following parts, since they tend to gather bran.

Table 10-1 Cleaning Place List

Parts to be cleaned periodically	Frequency
Outlet	Once a day
Bran discharge-way	Once a day
Chute	Once a week
Bran suction duct	Once a month
Screw feeder	Once a month

9.4 Inspection Item

CAUTION
• Be sure to lock out the motor before checking. (It shall take about 5 minutes for main shaft to stop completely after electric power is switched off.)

Item	Content	Frequency
Bearing failures (noise / heat)	• Run each equipment separately, and check for any unusual noise. • Check for any unusual heat at bearings after 10 minutes or more of continuous operation. • Replace the bearing if any abnormal conditions are found.	Once a month
V-belt tension	• The belt tension is judged appropriate when the belt deflects approximately 10 mm when pressed lightly with a finger.	Once a week
Screen wearing	• It can be used unit it breaks or deforms.	Every 3 months

Item	Content	Frequency
Milling roller Wearing	• Replace it when the diameter has been deceased by 3~5mm by wear.	Every 3 months
Resistance plate	• Replace when the resistance plate gets worn by about a half of its width.	Every 3 months

10. REPLACING CONSUMABLES

WARNING

- Before replacing the consumable parts, turn off the power switch and the breaker, and display a "**UNDER INSPECTION**" sign on the breaker.

10.1 Replacing the Screen

NOTE

- When replacing the screen with new one, remove oil or grease on the inside surface of the screen.

(1) Remove the cover.

(2) Remove only one frame.

(3) Remove retaining bolts of the inner frames.

(4) Remove the inner frames and the connecting bolts of the inner frames.

(5) Take out the inner frames one at a time.

(6) Remove the screen.

(7) Fit the new screen.

(8) Assemble the remained parts in reverse turn.

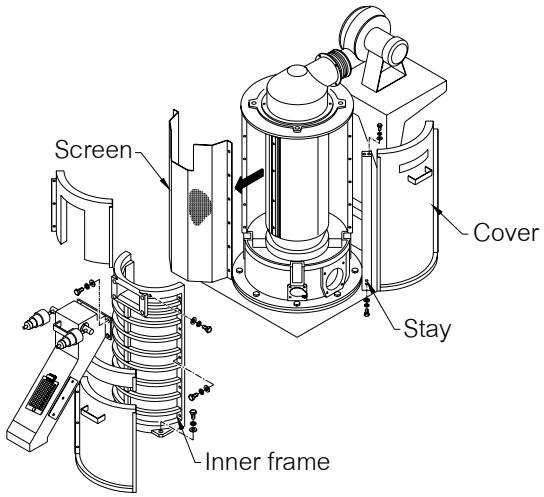


Fig. 10-1 Replacement of Screen

10.2 Replacing the Feed Roller and the Milling Roller

(1) Take the inner frames out of the machine by the procedure of No. (1)~(6) of section 10.1

(2) Loosen roller tightening nuts using a roller tightening tool, and raise the rollers. (The screw is right-thread.)

(3) The milling rollers are two-split type. Slightly raise the pieces for taking out one at a time.

(4) Fit the new milling rollers.

(5) Assemble the remained parts in reverse turn.

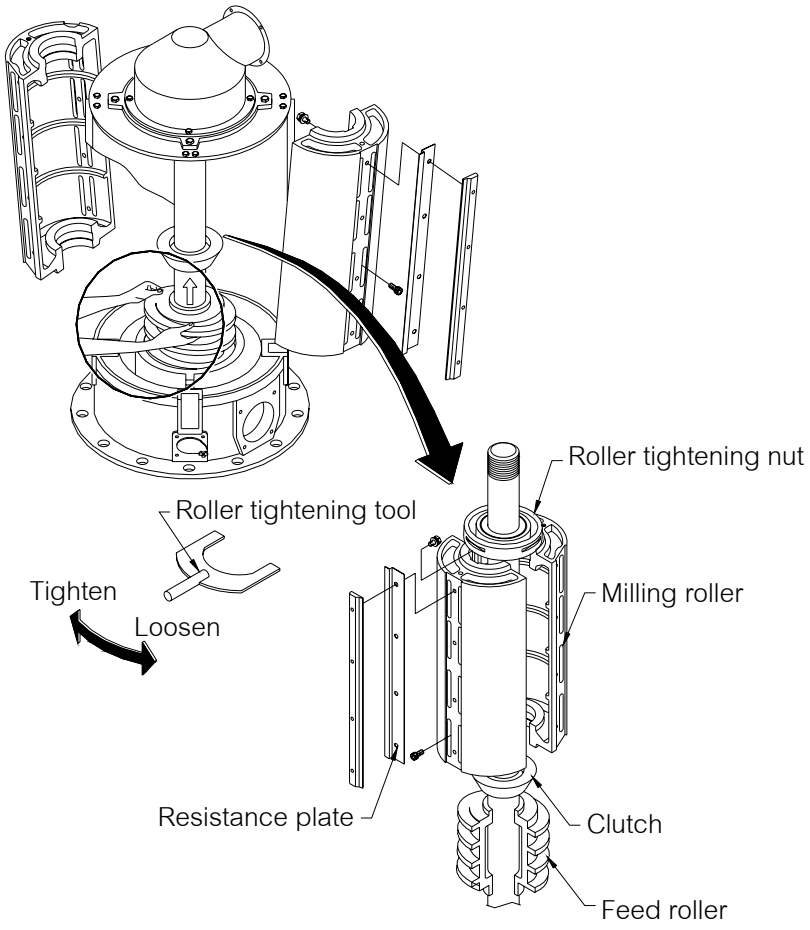


Fig. 10-2 Replacement of Feed Roller and Milling Roller

10.3 Attention in the Time of Assembling Milling Roller

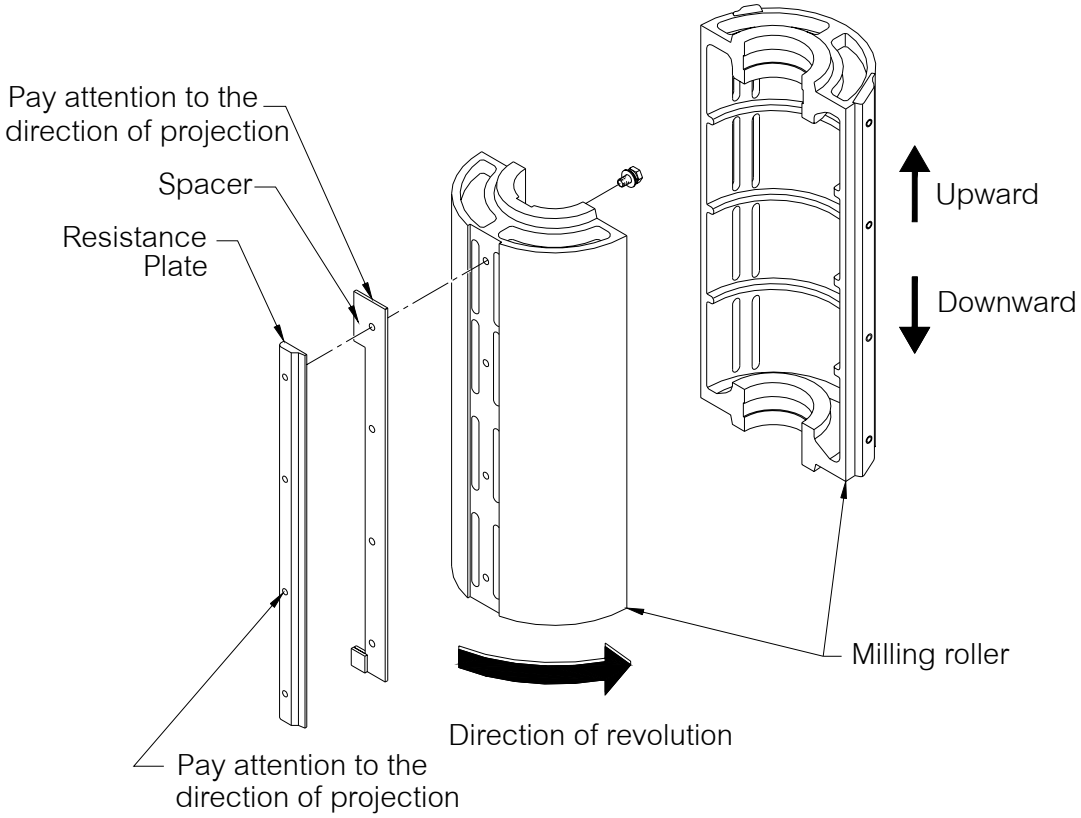
CAUTION

- Be careful not to assemble milling roller upside down because it can be assembled in both sides.
- If the assemble of milling roller is wrong, it cause the lower performance of machine such as increase of broken material, bad bran removal etc.

(1) Pay enough attention to the direction of milling roller when resistance plate and spacer are assembled. (Refer to figure below.)

(2) Pay also attention to the upward and downward direction when milling roller is assembled into machine shaft. (Refer to the figure below.)

Fig. 10-3 Assembling Procedure of Milling Roller



11. TROUBLESHOOTING

WARNING

- Before maintaining the machine, isolate the machine from the power supply.
- Refit the safety cover after maintenance.
- If the troubles are not remedied even after taking measures following the below instructions, contact your nearby distributor. See the contact address of Satake group companies at the end of this manual.

Conditions of Failure	Cause	Action to Take
1. Insufficient flow rate.	• Foreign matter in feed valve.	→ Remove the foreign matter.
	• Worn feed roller.	→ Replace the worn part.
	• Worn resistance plate.	→ Replace the worn plate.
2. A lot of broken material produced.	• Excessive flow rate.	→ Decrease the flow rate to reduce the load.
	• Excessive weight changed.	→ Decrease the load.
	• Worn resistance plate.	→ Replace the worn plate.
	• Worn screen.	→ Replace the worn part.
	• Worn milling roller.	→ Replace the worn part.
3. Excessive rise in the rice temperature.	• Insufficient air volume.	→ Increase the air volume.
	• Decrease efficiency by the wear of screen, milling roller etc.	→ Replace the worn part.
	• Adhesion of bran to the screen frame or the blinding.	→ Remove the bran stuck on the frame.

Conditions of Failure	Cause	Action to Take
4. Insufficient whiteness.	• Wear of screen, milling roller, etc. • Adhesion of bran to the screen frame or the blinding.	→ Replace the worn part. → Remove the bran stuck on the frame.
5. Bran insufficiently removed.	• Insufficient air volume.	→ Increase the air volume.